QUESTIONS AND SOLUTIONS

SECTION A: 5 marks each

1. Suppose $x + \frac{2}{y} = 2$ and $y + \frac{2}{x} = 4$. What is the value of $\frac{1}{2} + \frac{x}{y}$, for $x, y \neq 0$?

Solution

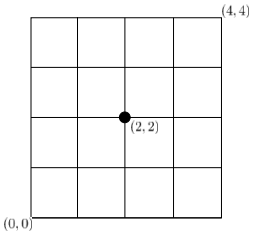
From $x + \frac{2}{y} = 2$ and $y + \frac{2}{x} = 4 \Leftrightarrow xy = 2y - 2 = 4x - 2 \Leftrightarrow \frac{x}{y} = \frac{1}{2}$ thus $\frac{x}{y} + \frac{1}{2} = 1$.

2. John started reading 16 books, one at a time. The first book took him one day to read, the second book took 2 days to read, the third took him three days to read, and so on. John finished reading the first book on a Monday and the second book on a Wednesday. On what day of the week did he finish reading the 16th book?

Solution

The number of days spent reading the 16 books is $1 + 2 + 3 + 4 + 5 + \ldots + 16 = 136$ days. The 16th day was read on 135th day after the 1st day (Monday) thus $135(\mod 7) = 2 \Rightarrow$ the day was Wednesday, denote remainders $0, 1, 2$ by $0-$ Monday, $1-$ Tuesday and $2-$ Wednesday.

3. A student moves from the point $(0, 0)$ to the point $(4, 4)$ in exactly eight steps each of distance 1 in either the positive x or y direction. What is the number of possible paths the student can follow so as he/she passes through the point $(2, 2)$?



Solution

There are two possible paths to leave $(0, 0)$, There are two possible paths to leave $(1, 0)$, There are two possible paths to leave $(0, 1)$, There are two possible paths to leave $(1, 1)$. Thus $2 \times 2 \times 2 \times 2 = 2^4 = 16$ paths passing through $(2, 2)$ plus two ways namely $(0, 0) \to (1, 0) \to (2, 0) \to (2, 1) \to (2, 2)$ and $(0, 0) \to (0, 1) \to (0, 2) \to (1, 2) \to (2, 2)$ in total 18 paths to $(4, 4)$ passing through $(2, 2)$.

4. In a supermarket there is a five day sales promotion where prices on all items are discounted by $\frac{1}{3}$ of the price on the first day. Beginning on the second day and each day after they have additional 10% off the previous day's price. What is the overall percentage discount that the buyer enjoys if he/she buys an item on the fourth day of the promotion?

Solution

The first day you pay $\frac{2}{3}x$, if x is the original price
The second day you pay $0.9 \times \frac{2}{3}x$
The third day you pay $0.9 \times 0.9 \times \frac{2}{3}x$
The fourth day you pay $0.9 \times 0.9 \times 0.9 \times \frac{2}{3}x = 0.486x$
Thus the overall percentage discount the buyer enjoys if he/she buys an item on the fourth day is 51.4%.

5. For every positive integer n, the Collatz function $f(n)$ is defined by;

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \in \mathbb{E}(\text{set of even numbers}) \\ 3n+1, & \text{if } n \in \mathbb{O}(\text{set of odd numbers}) \end{cases}$$

If $f^k(n) = f \circ f \circ f \circ \ldots \circ f(n)$, composed k-times in succession. Find the value of $f^{2023}(40)$.

Solution

$f(40) = 20, \quad f^2(40) = 10, \quad f^3(40) = 5, \quad f^4(40) = 16, \quad f^5(40) = 8, \; f^6(40) = 4,$
$f^7(40) = 2, \quad f^8(40) = 1, \quad f^9(40) = 4, \quad f^{10}(40) = 2, \quad f^{11}(40) = 1, \quad f^{12}(40) = 4$ and so on, the value of $f^k(40) = 4$ for $k > 8$ keeps repeating itself, now $f^{(8+3k)}(40) = 1 \;\; \forall k \;\in\; k = 0, 1, 2, 3, 4, \ldots$ so $2023 = 8 + 2015 = 8 + 671 \times 3 + 2$ so $f^{2023}(40) = f^{(8+671\times 3+2)}(40) = 2$.

6. What is the last digit of 3^{2023}?

Solution

Consider the following trial digits for a few powers, $3^1 = 3, \quad 3^2 = 9, \quad 3^3 = 27, \quad 3^4 = 81, \quad 3^5 = 243, \quad 3^6 = 729 \quad , 3^7 = 2187, \quad 3^8 = 6561, \quad 3^9 = 19683, \quad 3^{10} = 59049 \ldots$. There is a pattern in the last digit of $3, 9, 7, 1$ and repeats itself in the same order. 3^{4k-3} has the last digit as 3 for $k = 1, 2, 3, \ldots$, for $2023 = (4 \times 506) - 3 + 2 \Rightarrow 3^{2023}$ will have the last digit as 7.

7. A day is said to be prime if both the month and the day are prime. For example March 13th is a prime day since March (3) is prime and 13 is also a prime number. How many prime days are there in 2023?

Solution

The prime months are February (2), March (3), May (5), July (7), and November (11), the prime days are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29,$ and 31. Recall 2023 has 28 days in February, therefore Feb has 9 prime days, each of the months March, May and July has 11 prime days, November has 10 prime days (no 31st) $\Rightarrow$ a total of 52 prime days in 2023.

8. Among Juliet, David, Annet, Violet and Alice, there is exactly one spy. Each person makes one statement, the spy and exactly one other person will tell the truth. The statements are given in this order,
Juliet: Violet is not a spy
David: Violet is going to lie
Annet: The spy is either David or Violet
Violet: The spy has already made a statement
Alice: Iam the spy.
Who is the spy?

Solution

Juliet is the spy.

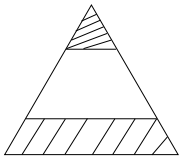
9. At the beginning of the first day, a box contains 1 black ball, 1 gold ball, and no other balls. At the end of each day, for each gold ball in the box, 2 black balls and 1 gold ball are added to the box; this means that at the end of the first day, there are 5 balls in the box. If no balls are removed from the box, how many balls are in the box at the end of the seventh day?

Solution

End of day	No. of Black balls	No. of Gold balls
1	$1+2$	$1+1$
2	$1+2+4$	$1+1+1+1$
3	$7+(4\times 2)$	$4+4$
4	$15+(8\times 2)$	$8+8$
5	$31+(16\times 2)$	$16+16$
6	$63+(32\times 2)$	$32+32$
7	$127+(64\times 2)$	$64+64=128$

The total number of balls is $128+255=383$ balls.

10. Consider a cake in the shape of an equilateral triangle with height 3 cm. Cut it into three strips of equal height 1 cm as shown in the figure below. Find the total combined area of the top and the bottom strips of the triangular cake.



Solution

The shaded triangle on top may be rotated and fit together with the shaded trapezoid on the bottom to form a parallelogram. The height of the parallelogram is 1 cm, we find the base of the parallelogram; the adjacent $=\frac{3}{\sqrt{3}}=\sqrt{3}$ the base $=2\times$ adjacent $\Rightarrow$ base $=2\sqrt{3}$, therefore the area of the parallelogram is $bh=2\sqrt{3}\times 1=2\sqrt{3}$ sq. units.

SECTION B : 10 marks each

11. Let x and y be positive numbers satisfying $2+\log_2 x=3+\log_3 y=\log_6(x+y)$. Find the value of $\frac{1}{x}+\frac{1}{y}$.

Solution

From $2+\log_2 x=3+\log_3 y=\log_6(x+y)$, Let $a=2+\log_2 x=3+\log_3 y=\log_6(x+y)$ then $x=2^{a-2}, y=3^{a-3}, x+y=6^a$ thus

$$\frac{1}{x}+\frac{x}{y}=\frac{x+y}{xy}=\frac{6^a}{2^{a-2}\times 3^{a-3}}=2^2\times 3^3=108.$$

12. The ratio of paint to water in three separate solutions of paint and water is $1:1, 1:2$ and $1:3$. If equal quantities of each solution are combined together and mixed what will be the ratio of water to paint in the resulting solution?

Solution

Assume that we take a bucket of each of the solution then solution 1 will contain $\frac{1}{2}$ water and $\frac{1}{2}$ paint, solution 2 will contain $\frac{2}{3}$ water and $\frac{1}{3}$ paint and solution 3 will contain $\frac{3}{4}$ water and $\frac{1}{4}$ paint, now we have, total fraction of paint as $\frac{1}{2}+\frac{1}{3}+\frac{1}{4}$ bucket of paint or $\frac{13}{12}$ paint. For water we have $\frac{1}{2}+\frac{2}{3}+\frac{3}{4}$ buckets or $\frac{23}{12}$ thus the required ratio is $23:13$.

13. The distance between the dormitory and the class is $1KM$. You leave the dormitory at the same time each day, walk at a steady speed of 3 KM per hour and arrive just as the class starts. Today you were distracted by the good weather and walked the first half kilometre at a speed of 2 KM/hr. At what speed must you run(walk) the last half of the kilometre in order to arrive just as the class starts?

Solution

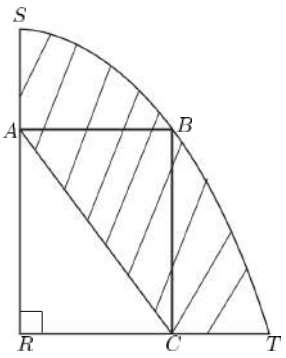
Recall Distance = speed $\times$ time. Initially you take $t = \frac{2}{3}$ hours = 20 minutes from dormitory to class. Today it took you 15 minutes to cover first half KM i.e Distance = speed $\times$ time thus $\frac{1}{2} = 2\times$ time $\Rightarrow t = 15$ minutes. You therefore have 5 minutes to reach on time from $\frac{1}{2}$ KM = speed $\times$ time where $t = 5mins = \frac{1}{12}hr$ thus speed = 6 km/hr.

14. What is the value of $\log_3 4 \times \log_4 5 \times \log_5 6 \times \ldots \times \log_{80} 81$?

Solution

Convert all logarithms into base 10 that is,
$\log_3 4 \times \log_4 5 \times \log_5 6 \times \ldots \times \log_{80} 81 = \frac{\log_{10} 4}{\log_{10} 3} \times \frac{\log_{10} 5}{\log_{10} 4} \times \frac{\log_{10} 6}{\log_{10} 5} \times \cdots \frac{\log_{10} 81}{\log_{10} 80} = \frac{\log_{10} 81}{\log_{10} 3} = 4.$

15. In the figure on the right, arc SBT is one quarter of a circle with centre R and radius r. The length plus width of the rectangle ABCR is 8 units and the perimeter of the shaded region is $10 + 3\pi$. Find the value of r.



Solution

Let $x = \overline{RC}, y = \overline{AR} \Rightarrow x+y = 8$, Now $x^2+y^2 = r^2$ since the rectangle $ABCR$ has one vertex in the circle. The perimeter of the shaded area is $AS+SBT+TC+AC = (r-y)+\frac{\pi r}{2}+(r-x)+\sqrt{x^2+y^2} = 3r + \frac{\pi r}{2} - 8$ thus $18 + 3\pi = (3 + \frac{\pi}{2})r \Rightarrow r = 6$ units.